Earned Value Management & Project Status Reporting

Agenda

Chapter 13 Problems

- Exercise 3: Status Report (End of Period 4 & 8)
- Exercise 4 : Multi-Period Status
- Exercise 7 : 12-Period Analysis & Graphing
- Case 13.3: Scanner Project Narrative Report

Chapter 13 | Exercise 3 - Introduction

Goal: Develop a status report for periods 4 and 8 using Earned Value metrics.

Project baseline (PV) (in $)																				
Task	DUR	ES	LF	SL	Budget (PV)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	2	0	2	0	400	200	200														
B	6	2	8	0	2400			200	600	200	600	200	600								
C	5	2	9	2	1500			200	400	500	100	300									
D	4	8	12	0	1600									400	400	400	400				
E	3	7	12	2	900								300	400	200						
F	3	12	15	0	600													200	100	300	
Period PV total						200	200	400	1000	700	700	500	900	800	600	400	400	200	100	300	
Cumulative PV total						200	400	800	1800	2500	3200	3700	4600	5400	6000	6400	6800	7000	7100	7400	

Chapter 13 | Exercise 3 - Introduction

Key Formulas

- **EV (Earned Value)** = % complete x Budget
- **PV (Planned Value)** = Budgeted value of work **scheduled** to be done by that time
- **AC (Actual Cost)** = Actual money spent (given)
- **CV (Cost Variance)** = EV – AC
- **SV (Schedule Variance)** = EV – PV
- **CPI (Cost Performance Index)** = EV / AC
- **SPI (Schedule Performance Index)** = EV / PV

Baseline:

Budgets (BAC) per task:

- A = $400
- B = $2,400
- C = $1,500
- D = $1,600
- E = $900
- F = $600

Total BAC = 400 + 2400 + 1500 + 1600 + 900 + 600 = $7,400

Chapter 13 | Exercise 3 - Status at End of Period 4

EV = **Total EV_4** = 400 + 1200 + 500 + 0 + 0 = **$2,100**
PV = 400 + 800 + 600 = **$1,800**

CV = $EV_4 - AC_4$ = 2100 − 1800 = **+300**

SV = $EV_4 - PV_4$ = 2100 − 1800 = **+300**

CPI = EV_4 / AC_4 = 2100 / 1800 ≈ **1.17**

SPI = EV_4 / PV_4 = 2100 / 1800 ≈ **1.17**

Interpretation: Project is under budget and ahead of schedule as CV and SV >1

Chapter 13 | Exercise 3 - Status at End of Period 8

EV = $\mathbf{EV_8}$ = 400 + 2400 + 1500 + 400 + 300 + 0 = **$5,000**
PV = $4,600

CV = $\mathbf{CV_8(total)}$ = $EV_8 - AC_8$ = 5000 − 4600 = **+400**

SV = $EV_8 - PV_8$ = 5000 − 4600 = **+400**

CPI = EV_8 / AC_8 = 5000 / 4600 ≈ **1.09**

SPI = EV_8 / PV_8 = 5000 / 4600 ≈ **1.09**

Forecast: Expected to finish under budget and ahead of schedule.

Chapter 13 | Exercise 3 - Form

End of Period 4

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	400	300	400	100	0
B	50%	1100	1000	800	300	400
C	33%	500	500	600	0	-100
D	0%	0	0	0	0	0
E	0%	0	0	0	0	0
Cumulative Totals		2400	1800	1800	500	500

End of Period 8

Task	Actual % Complete	EV	AC	PV	CV	SV
A	Finished	400	300	400	100	0
B	Finished	2400	2200	2400	200	0
C	Finished	1500	1500	1500	0	0
D	25%	400	300	0	100	400
E	33%	500	300	500	0	0
F	0%	0	0	0	0	0
Cumulative Totals		5000	4600	4600	400	400

Chapter 13 | Exercise 3 - Conclusion

What information are you prepared to tell the customer about the status of the project at the end of period 8?

Key Activities:

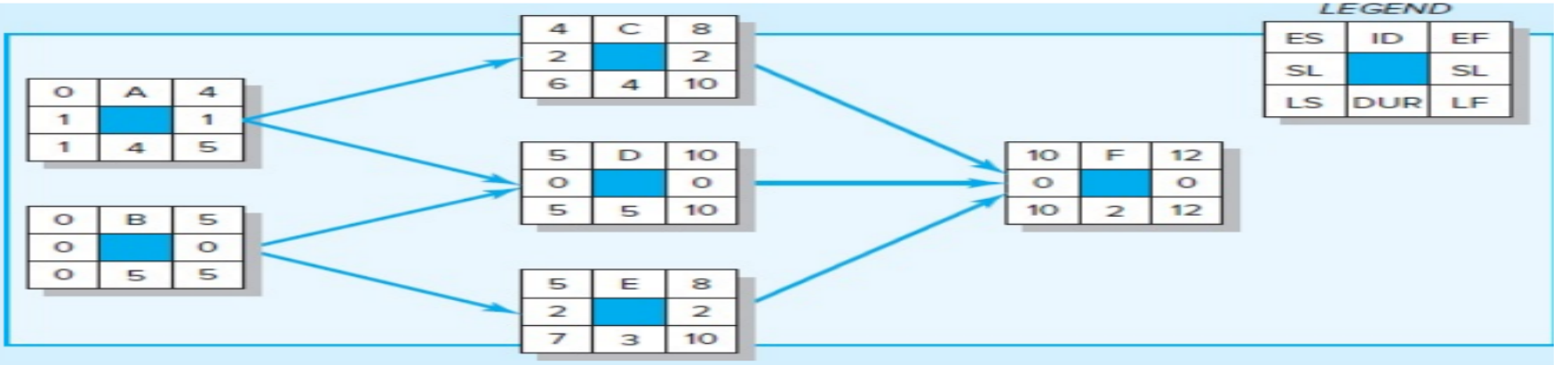
- A, B, C: Completed on/under budget.
- D: Started earlier than planned → creates schedule buffer.
- E: Progress aligns with plan.
- F: Not yet started and not required by period 8.

Outlook:

- Forecast **EAC ≈ $6,790** vs **BAC = $7,400** → projected **under budget and on/ahead of schedule** if current performance holds.

Chapter 13 | Exercise 4 - Introduction

Goal: Develop status reports for periods 2, 4, 6, and 8.



ID	Budget ($000)	0–1	1–2	2–3	3–4	4–5	5–6	6–7	7–8	8–9	9–10	10–11	11–12
A	40	10	10	10	10								
B	32	8	4	8	4	8							
C	48					12	12	12	12				
D	18						6	2	2	2	6		
E	28						8	8	12				
F	40											20	20
Total	206	18	14	18	14	20	26	22	26	2	6	20	20
Cumulative		18	32	50	64	84	110	132	158	160	166	186	206

Chapter 13 | Exercise 4 - Status at Period 2 & 4

Budgets (BAC):

A = 40

B = 32

C = 48

D = 18

E = 28

F = 40

Key Formulas:

PCIB = EV / BAC(total) (Percent Complete Index – Budget)

Chapter 13 | Exercise 4 - Performance Summary Table

	EV	AC	PV	SPI	CPI	PCIB
2	46	37	32	1.44	1.24	0.22
4	72	59	64	1.13	1.22	0.35
6	122	93	110	1.11	1.31	0.59
8	154	131	158	0.97	1.18	0.75

Chapter 13 | Exercise 4 - Assessment & Forecast

We're asked for **EAC(f)** and **VAC(f)**, using cumulative **CPI at period 8**:

$CPI_8 \approx 154 / 131 \approx$ **1.18**

EAC(f) (forecasted cost at completion using CPI):

$$\text{aa} \quad \text{a}_{\alpha} = \frac{\text{aaa}}{\text{aƎ} \quad \text{a}_{8}} = \frac{206}{154/131} \approx 175.2$$

VAC(f) (variance at completion forecast):

$$\text{ƎA} \quad \text{a}_{\alpha} = \text{aaa} \quad - \text{aa} \quad \text{a}_{\alpha} \approx 206 - 175.2 \approx + 30.8$$

So the project is forecast to finish about **$30.8K under budget**.

Conclusion: Strong cost control, slightly behind schedule.

Chapter 13 | Exercise 4 - Conclusion

Based on your data, what is your assessment of the current status of the project? At completion?

Forecast

- **BAC = 206**, **EAC ≈ 175**, **VAC ≈ +31** → projecting to finish **~$31K under budget**.

With SPI close to 1 and major tasks completed, project is expected to **finish roughly on time** if D & F are managed closely.

Schedule

- Slightly behind → **PV = 158**, **SV = −4**, **SPI ≈ 0.97** (minor slippage driven by Activity D).

Chapter 13 | Exercise 4 - Form

Status Report: Ending Period 2 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	75%	30	25	20	5	10
B	50%	16	12	12	4	4
Cumulative Totals		46	37	32	9	14

Status Report: Ending Period 4 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	24	8	8
Cumulative Totals		72	59	64	13	8

Status Report: Ending Period 6 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	32	8	0
C	75%	36	24	24	12	12
D	0%	0	0	6	0	-6
E	50%	14	10	8	4	6
Cumulative Totals		122	93	110	29	12

Status Report: Ending Period 8 ($000)

Task	% Complete	EV	AC	PV	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	32	8	0
C	100%	48	32	48	16	0
D	33%	6	20	10	-14	-4
E	100%	28	20	28	8	0
Cumulative Totals		154	131	158	23	4

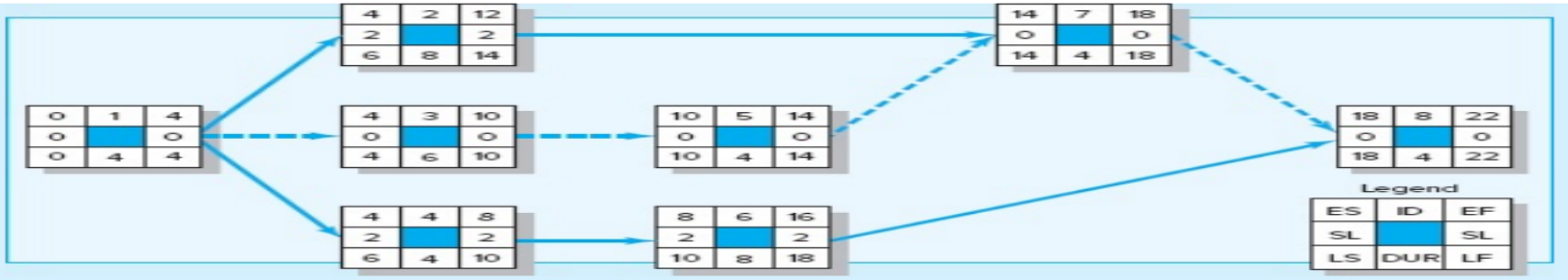
Performance Indexes Summary

Period	EV	AC	PV	SPI	CPI	PCIB
2	46	37	32	1.44	1.24	0.22
4	72	59	64	1.13	1.22	0.35
6	122	93	110	1.11	1.31	0.59
8	154	131	158	0.97	1.18	0.75

EAC_t = 175 VAC_t = 31

Chapter 13 | Exercise 7 - Introduction

Goal: Compute SV, CV, SPI, and CPI for periods 2-12 and assess project at period 12.



Task	DUR	ES	LF	SL	PV ($00)	Baseline (PV) ($00) 0–2	2–4	4–6	6–8	8–10	10–12	12–14	14–16	16–18	18–20	20–22
1	4	0	4	0	8	4	4									
2	8	4	14	2	40			10	10	10	10					
3	6	4	10	0	30			10	15	5						
4	4	4	10	2	20			10	10							
5	4	10	14	0	40						20	20				
6	8	8	18	2	60					20	20	10	10			
7	4	14	18	0	20								10	10		
8	4	18	22	0	30										20	10
Period PV total						4	4	30	35	35	50	30	20	10	20	10
Cumulative PV total						4	8	38	73	108	158	188	208	218	238	248

Status Report: Ending Period 2 **($00)**

Task	%Complete	EV	AC	PV	CV	SV
1	50%	—	4	—	—	—
Cumulative Totals		—	**4**	—	—	—

Chapter 13 | Exercise 7 - Introduction

Budgets (from PV column):

Task 1 = 8

Task 2 = 40

Task 3 = 30

Task 4 = 20

Task 5 = 40

Task 6 = 60

Task 7 = 20

Task 8 = 30

BAC = 8 + 40 + 30 + 20 + 40 + 60 + 20 + 30 = 248

Chapter 13 | Exercise 7 - Performance Index Summary

Period	SPI	CPI	PCIB
2	1.00	1.00	0.02
4	1.00	0.80	0.03
6	0.74	0.76	0.11
8	0.52	0.69	0.15
10	0.83	0.73	0.36
12	0.94	0.70	0.60

Chapter 13 | Exercise 7 - Assessment & Forecast

EAC(f) (forecast cost if current cost efficiency continues):

$$\text{aa a}_{\alpha} = \frac{\text{aaa}}{\text{aƎ a}_{12}} = \frac{248}{148/210} = \frac{248 \times 210}{148} \approx 352$$

VAC(f) (variance at completion):

$$\text{Ǝa a}_{\alpha} = \text{aaa} - \text{aa a}_{\alpha} \approx 248 - 352 = -104$$

So the project is trending about **$10,400 over budget**.

Conclusion: Project trending to finish $10,400 over budget.

Chapter 13 | Exercise 7 - Form

Status Report: Ending Period 2 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	50%	4	4	4	0	0
Cumulative Totals		4	4	4	0	0

Period	SPI	CPI	PCIB
2	1	1	0.02
4	1	0.8	0.03
6	0.74	0.76	0.11
8	0.82	0.69	0.15
10	0.83	0.73	0.36
12	0.74	0.7	0.60

SPI = EV/PV

CPI = EV/AC

PCIB = EV/BAC

Chapter 13 | Exercise 7 - Form

Status Report: Ending Period 4 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
Cumulative Totals		8	10	8	-2	0

Status Report: Ending Period 6 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	25%	10	15	10	-5	0
3	33%	10	12	10	-2	0
4	0%	0	0	10	0	-10
Cumulative Totals		28	37	38	-9	-10

Status Report: Ending Period 8 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	30%	12	20	20	-8	-8
3	60%	18	25	25	-7	-7
4	0%	0	0	20	0	-20
Cumulative Totals		38	55	73	-17	-35

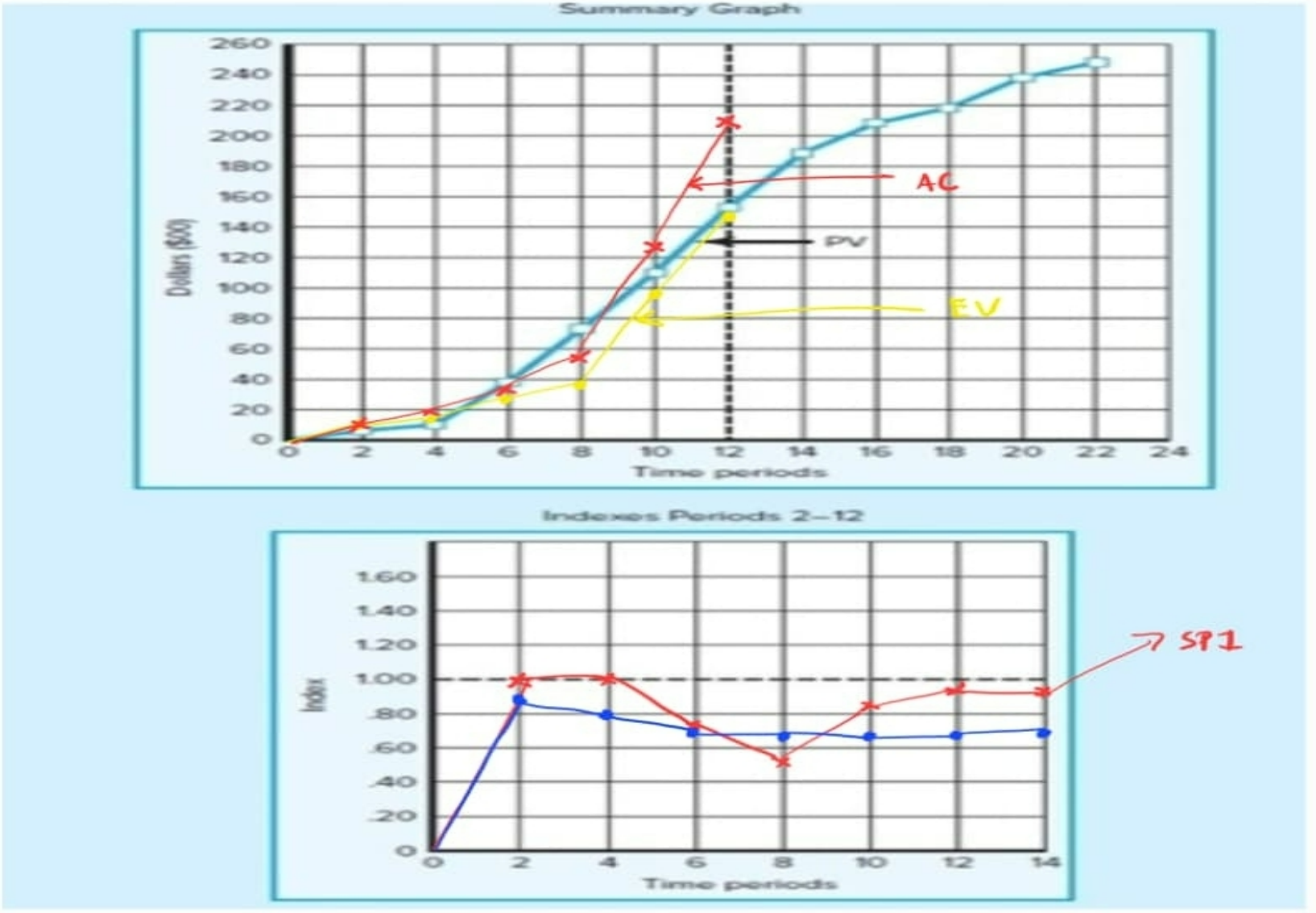
Status Report: Ending Period 10 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	60%	24	30	30	-6	-6
3	Finished	30	40	30	-10	0
4	50%	10	20	20	-10	-10
5	0%	0	0	0	0	0
6	30%	18	24	20	-6	-2
Cumulative Totals		90	124	108	-34	-18

Status Report: Ending Period 12 ($00)

Task	%Complete	EV	AC	PV	CV	SV
1	Finished	8	10	8	-2	0
2	Finished	40	50	40	-10	0
3	Finished	30	40	30	-10	0
4	Finished	20	40	20	-20	0
5	50%	20	30	20	-10	0
6	50%	30	40	40	-10	-10
Cumulative Totals		148	210	158	-62	-10

Chapter 13 | Exercise 7 - Graph



Case 13.3 - Scanner Project Overview

Goal: Provide a non-technical status report for the board.

Electroscan, Inc.
555 Acorn Street, Suite 5
Boston, Massachusetts

29 In-store Scanner Project
(thousands of dollars)
Actual Progress as of January 1

Name	PV	EV	AC	SV	CV	BAC	EAC_f
Scanner project	420	395	476	−25	−81	915	1103
H 1.0 Hardware	92	88	72	−4	16	260	213
H 1.1 Hardware specifications (DS)	20	20	15	0	5	20	15
H 1.2 Hardware design (DS)	30	30	25	0	5	30	25
H 1.3 Hardware documentation (DOC)	10	6	5	−4	1	10	8
H 1.4 Prototypes (PD)	2	2	2	0	0	40	40
H 1.5 Test prototypes (T)	0	0	0	0	0	30	30
H 1.6 Order circuit boards (PD)	30	30	25	0	5	30	25
H 1.7 Preproduction models (PD)	0	0	0	0	0	100	100
OP 1.0 Operating system	195	150	196	−45	−46	330	431
OP 1.1 Kernel specifications (DS)	20	20	15	0	5	20	15
OP 1.2 Drivers	45	55	76	10	−21	70	97
OP 1.2.1 Disk drivers (DEV)	25	30	45	5	−15	40	60
OP 1.2.2 I/O drivers (DEV)	20	25	31	5	−6	30	37
OP 1.3 Code software	130	75	105	−55	−30	240	336
OP 1.3.1 Code software (C)	30	20	40	−10	−20	100	200
OP 1.3.2 Document software (DOC)	45	30	25	−15	5	50	42
OP 1.3.3 Code interfaces (C)	55	25	40	−30	−15	60	96
OP 1.3.4 Beta test software (T)	0	0	0	0	0	30	30
U 1.0 Utilities	87	108	148	21	−40	200	274
U 1.1 Utilities specifications (DS)	20	20	15	0	5	20	15
U 1.2 Routine utilities (DEV)	20	20	35	0	−15	20	35
U 1.3 Complex utilities (DEV)	30	60	90	30	−30	100	150
U 1.4 Utilities documentation (DOC)	17	8	8	−9	0	20	20
U 1.5 Beta test utilities (T)	0	0	0	0	0	40	40
S 1.0 System integration	46	49	60	3	−11	125	153
S 1.1 Architecture decisions (DS)	9	9	7	0	2	10	8
S 1.2 Integration hard/soft (DEV)	25	30	45	5	−15	50	75
S 1.3 System hard/software test (T)	0	0	0	0	0	20	20
S 1.4 Project documentation (DOC)	12	10	8	−2	2	15	12
S 1.5 Integration acceptance testing (T)	0	0	0	0	0	30	30

Case 13.3 - Insights

- The project has **EV = $395K**, **PV = $420K**, **AC = $476K** → **behind schedule (SV = −$25K, SPI = 0.94)** and **over budget (CV = −$81K, CPI = 0.83)**.

- If current performance continues, **forecast final cost ≈ $1.10M vs BAC $915K** → projected **overrun ≈ $188K (~21%)**.

- **Hardware**: On schedule and under budget → low risk.

- **Operating System**: Behind schedule and overspending → **primary driver** of current and future overrun.

- **Utilities & System Integration**: Progressing, but at **poor cost efficiency**, adding to overruns.

Case 13.3 – Recommendations

- Focus corrective action on OS & Utilities
- Freeze nonessential changes
- Tighten scope
- Shorten milestones
- Continue **monthly EV reviews**.

Outlook

- Project can deliver required functionality, but **cost risk is high**; timely intervention is essential to protect budget and schedule.